

Sivan Syed

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Austin, TX

🌐 <https://ilgneous.github.io/>

Education

The University of Texas at Austin (GPA: 3.8/4.0)

May 2027

B.S. Electrical Engineering, B.S. Physics, B.S. Mathematics, Certificate German

Relevant Coursework: Stochastic Processes, Machine Learning, Automatic Control Systems, Embedded Systems

Graduate Coursework: Computational Physics, Magnetic Materials and Devices, Quantum Mechanics I

Experience

Emerson – Electrical Engineering Design Intern

May 2026–Aug 2026

- Modeled a 4–20 mA op-amp and potentiometer signal-conditioning schematic enabling field calibration of set-points of a reverse-engineered raw production PCB
- Troubleshoot and debugged electrical systems, such as power electronics and control systems to resolve performance or safety issues
- Collaborated with Production, Manufacturing, and Engineering teams to author product Work Instructions
- Proposed and spearheaded the restructuring of Magtech data organization, improving query speed

Adom Industries Inc. – R&D Design Engineer Intern

May 2025–Aug 2025

- Architected a mixed-signal Vehicle Control Unit (VCU) utilizing an RP5 and RP2040 to bridge low-level CAN bus motor control with high-level ROS path planning
- Designed and routed custom mixed-signal PCBs for LiDAR and motor control, implementing digital isolation to successfully mitigate electromagnetic interference (EMI) in a 48V power system
- Applied iterative FEA to analyze stress, strain, and deformation, reducing material usage 88%

Sandia National Laboratories – R&D Computational Engineer

May 2024–Aug 2024

- Built a fast current-pulse generator modeling capacitor discharges, synthesizing 10k+ waveforms for ML surrogate modeling and Uncertainty Quantification
- Trained an FNN surrogate model replicating magneto-hydrodynamics trajectory simulations at >98% fidelity
- Analyzed fluid RTIs and RMIIs to diagnose simulation inaccuracies
- Wrote ODE trajectory solvers to track radial liner implosions and compare observed instability spectra

Projects and Organizations

FSAE Longhorn Racing – Dynamics + Vehicle Modeling Engineer, Steering Lead

Sep 2022–Aug 2025

- Analyzed and tuned kinematic and tire models (MF combined slip) in a Model-in-the-Loop simulation, validating tire and slip-angle targets against track telemetry
- Developed a compliance model using first-principles physics to simulate part stress with FEM collision detection
- Owned, designed, and manufactured the steering system from steel to a carbon-fiber layup with a metal 3D-printed bevel gearbox, cutting weight 68% at equal torsional rigidity

Edge-ML UAV Flight Controller

- Designing an ML-integrated flight controller with a nonlinear Kalman filter fusing noisy IMU measurements
- Developed a 10-DoF quadcopter dynamics simulator, implementing unit and regression testing suites to validate first-principles physics and guarantee robust model fidelity
- Architected deterministic flight control firmware in Rust utilizing RTIC (Real-Time Interrupt-driven Concurrency); applied Rate Monotonic Scheduling (RMS) to safely isolate safety-critical attitude loops from ML inference tasks

SOT-Driven Skyrmion Racetrack Memory Device

- Designed and tested a novel SOT-driven magnetic skyrmion racetrack memory in MuMax3, using a passive defect channel and VCMA gate to pin, release, and steer skyrmions on demand via SLURM

Time-Series Damage Forecasting

- Forecast damage with an LSTM at >98% accuracy for optimal insurance pricing, benchmarking ARIMA, LSTM, and Transformer models head-to-head

Skills

Languages: C, C++, ARM, Rust, Python, MATLAB, Bash, JS, HTML, CSS, LaTeX

Tools: KiCAD, Cadence, Git/GitHub, Draftsight, LTSpice, MuMax3, Fusion360, SOLIDWORKS, Ansys